1. The curve C has equation

$$y = (3x - 2)^6$$

(a) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$

(2)

Given that the point $P\left(\dfrac{1}{3}, 1\right)$ lies on C,

(b) find the equation of the normal to C at P. Write your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)